

Aleksandr Razin

TBILISI · GEORGIA

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Research Scientist with industrial background (~5 years): applied research at Huawei, production ML engineering, independent research. First & corresponding author of an ECCV 2026 paper on diffusion latent upscaling; led quantized post-training of night-HDR video restoration shipped on Huawei devices; built a multi-agent multimodal AI copilot for 2M+ monthly users; founded a computer-vision startup (acquired 2025). Research interests: **visual content generation & understanding, multimodal agents, robot learning**. M.Sc. ML (ITMO); INSAIT & ELLIS PhD student.

Experience

Independent Researcher

Nov 2025 – Present

- First & corresponding author of **Latent Upscaling Adapter** (accepted **ECCV 2026**); led theory and experiment design

TradingView

SENIOR ML ENGINEER

Sep 2025 – Present

- Engineered the harness and full offline/online evaluation for an agentic **AI copilot** (thousands of tools, hundreds of agents) for real-time chart analysis & modification; **2M+ monthly users**
- Built production **multimodal** search (text, charts/images, video) on fine-tuned embeddings + hybrid retrieval; **~80% precision** per modality
- Cut existing LLM-solution costs **~20%/month** by blending third-party APIs with local inference

Startup

FOUNDER, LEAD ML ENGINEER

Apr 2025 – Sep 2025

- Founded and hired a **5-engineer** product team; secured ~€100K; **acquired (2025)** by a large IT company
- Shipped a retail **video-analytics** CV suite (detection, segmentation & tracking) to 3 of top-5 national retailers, saving ~€20K+/month
- Post-trained **text-to-image / image-to-image** ad-generation models, improving identity preservation and lifting client ad **CTR ~15%**

Huawei Research Institute

ARTIFICIAL INTELLIGENCE RESEARCHER

Dec 2023 – Apr 2025

- Redesigned **video restoration** architectures (attention + convolution optimization), improving perceptual quality, detail reconstruction & color
- Optimized **diffusion training** (checkpoint-merging & loss design) for high-frequency texture synthesis; **~10%** gains over baselines (FID/PSNR)
- Led **quantized post-training** for the **Mate 70** night-HDR video restoration network (enhancement, super-resolution, joint denoising & demosaicing); refined strategy cut ghosting & flicker, shipping weekly to production

RESEARCH INTERN

May 2023 – Sep 2023

- Improved domain **LLM** generation > **30% ROUGE** via classification-agent query routing and LoRA + DPO post-training
- Enabled **multimodal** LLM agents by training metric-learned visual encoders on 100K+ sample datasets for visual question answering

ITMO University | Industry AI Laboratory

DEEP LEARNING ENGINEER

Jun 2022 – May 2023

- Built an end-to-end **data pipeline** (extraction, labeling, processing, analysis), raising detection & segmentation Precision/Recall
- Distilled banknote segmentation models (**20% fewer FLOPs**, no accuracy loss); **contrastive learning** lifted fake-stamp detection +10% F1

Mining University | Robotic Laboratory

RESEARCH INTERN

Jan 2021 – Dec 2021

- Built **on-device** software and trained lightweight models (detection, classification, segmentation) for obstacle & defect detection on **pipeline robots** at oil & gas plants, cutting manual inspection by ~€56K+/year

Education

INSAIT, Sofia University – ETH, Zurich

PHD MACHINE LEARNING

2026 – Present

Advisors: Dr. Jinjin Gu (INSAIT), Prof. Marc Pollefeys (ETH)

ITMO University

M.Sc. MACHINE LEARNING, GPA: 4.9/5 – THESIS: *Enhancing Image Super-Resolution through Depth Estimation*

2022 – 2024

Teaching Assistant: *Advanced Machine Learning Technologies*

Mining University

B.Sc. ENGINEERING, GPA: 4.9/5 – THESIS: *Real-time obstacle detection and analysis for robotic exploration with LiDAR*

2018 – 2022

Teaching Assistant: *Experimental Physics*

Mathematics Tutor (Volunteer): *Olympiad mathematics & competitive programming*

Selected publications

2026 **A. Razin**, D. Kazantsev, I. Makarov. *One Small Step in Latent, One Giant Leap for Pixels: Fast Latent Upscale Adapter for Your Diffusion Models*. Hugging Face Daily Paper #1

ECCV 2026

2024

Gorbatosvski, A. V., **Razin, A. D.**, Aliev, A. A., Kovalchuk, S. V. *Improving question answering in programming domain with pretrained language model fine-tuning using structured diverse online forum data.* Journal of ITMO

Journal

2022

Psheninin, V., Liagova, A., **Razin, A.**, Skorobogatov, A., Komarovsky, M. *Robot crawler for surveying pipelines and metal structures of complex spatial configuration.* MDPI Infrastructures

Journal

Skills

Languages Python, Go, JAX (basic)
Tools Docker, Kubernetes, Kafka, AWS, Git, CI/CD, CUDA Toolkit, MLflow, Airflow, Kubeflow
Frameworks PyTorch, TorchVision, OpenCV, Transformers, Diffusers, DeepSpeed, DiffSynth-Studio, ComfyUI, vLLM, Google ADK
Concepts Vision / Language / Multimodal models, 3D vision, foundation models, distributed training, model & training optimization, post-training, experiment management, A/B testing, MLOps, SWE
Language Skills English (fluent), Russian (native)

Awards

ELLIS PhD Program: shortlisted & offered a PhD (top 15% of 6,000+ applicants) 2026
Huawei Internship: after successful participation, offered a full-time position as a CV Engineer 2023
Kaggle: “OTTO – Multi-Objective Recommender System”: top 10% out of 2,574 teams 2023
Grant winner, Russian Student Startup: top 5% out of ~10k teams (~€15k funding) 2023
Prize holder, Russian Olympiad in Phys. & Math. “Star”: Spacecraft track, top 2% of ~8k participants 2018
School Gold medal: excellent academic performance at high school 2018